

Write-up: Model Documentation

1. Who are you?

I am **ChengHuige**, a software engineer based in Beijing, China.

I have participated in many Kaggle competitions and am a **Kaggle Grandmaster**, with a peak ranking of **#7** worldwide. I have also competed in and won numerous domestic competitions in China, including **1st place in the Tencent WBDC 2021** (We-Chat Big Data Challenge).

For this challenge, all source code was authored with the assistance of GPT-4 and Claude (Copilot coding agents); I focused on experiment design, strategy, ablation analysis, and iteration direction.

2. Motivation

Children's ASR is a challenging and impactful problem — child speech is highly variable (age, pronunciation development, L1 transfer). The phonetic track (IPA transcription) adds extra difficulty: models must output fine-grained phonetic sequences, not just words. I was drawn to the unique combination of ASR + phonetics + low-resource child speech.

3. High-Level Approach

11-model heterogeneous ensemble + CatBoost tree reranker:

Component	Details
NeMo Parakeet-TDT-0.6B (3 TDT + 4 CTC)	Main backbone. TDT converges fast (~5 epochs), excels on long audio & DD data
WavLM-Large (4 CTC models)	Excels on short audio & EXT (TalkBank) data. 5h/epoch on Pro 6000
Dual-head training	IPA CTC/TDT head + Word BPE CTC head (weight 0.3) — leverages larger word-track data
Concat Mix augmentation	Concatenate up to 8 audio clips + labels; most impactful single augmentation
CatBoost LambdaRank reranker	5-fold CV, ~200 features, selects best N-best candidate per utterance

None

Raw Audio → 11 Models (N-best) → CTC Rescoring → CatBoost Reranker → Final IPA

Key insight: WavLM excels on short/EXT audio, NeMo TDT excels on long/DD audio — this complementarity is the foundation of the ensemble.

4. Visualizations

Single model CV (fold 0, IPA CER ↓):

Model	Overall	DD	EXT
v17.wavlm-large.ep3.5	0.2900	0.3445	0.2355
v16.wavlm-large.dual_bpe.mix4	0.2923	0.3477	0.2368
v16.dual_bpe.mix2.mix_csss.tdt_only	0.2913	0.3416	0.2410
v16.dual_bpe.tdt_only	0.2931	0.3447	0.2415
v16.dual_bpe.mix4	0.2928	0.3393	0.2464
v16.aux_loss.dual_bpe	0.2952	0.3418	0.2486

Ensemble CV (fold 0):

Method	CER
Best single model	0.2900
Baseline (best-of-models per utt)	0.2724
Full-avg MBR	0.2672
CatBoost Reranker (final)	0.2628
Oracle (upper bound)	0.1685

5. Three Most Impactful Code Segments

5.1 Concat Mix Augmentation (`dataset.py`)

Python

```
# Concatenate multiple audio clips + labels for data augmentation
# Most impactful single augmentation technique
def _concat_mix(self, batch_items, max_mix=8):
    """Randomly concat up to max_mix audio segments with their IPA labels.
    Strategies:
    - 'csss': select from DD/EXT with equal probability (DD-friendly)
    - 'random': uniform random from DD+EXT combined pool
    - fit_cost: limit total cost = audio_sec * label_units <= max_cost(120)
    """
    mixed_audio = []
    mixed_labels = []
    for item in selected_items:
        mixed_audio.append(item['audio'])
        mixed_labels.append(item['label'])
    return torch.cat(mixed_audio), ' '.join(mixed_labels)
```

Impact: Child speech utterances are very short (1-3 words). Concatenating clips creates longer, more diverse training examples. No-mix → mix8 improved single-model CER by ~0.01 absolute.

5.2 Dual-Head Training (`models/nemo.py`)

Python

```
# Two decoder heads sharing one encoder
# Primary: IPA phonetic CTC/TDT (vocab=53)
# Secondary: Word BPE CTC (weight=0.3) - leverages larger word-track data
enc_out = self.encoder(audio)
phon_loss = self.phonetic_head(enc_out, phonetic_labels)
word_loss = self.word_bpe_head(enc_out, word_labels)
total_loss = phon_loss + 0.3 * word_loss
```

Impact: Word track has ~5x more training data. Shared encoder benefits from this larger dataset. Improved local CV by ~0.003 absolute.

5.3 CatBoost Tree Reranker (`ensemble.py` + `ensemble_feats.py`)

Python

```
# Build feature matrix: CTC scores (per model), edit distances, audio features
features = build_reranker_features(candidates_df, ctc_scores, model_names)
# CatBoost LambdaRank selects best candidate per utterance
```

```
catboost_model = CatBoostRanker(
    iterations=1000, learning_rate=0.05, depth=6,
    loss_function='YetiRankPairwise',
)
catboost_model.fit(X_train, y_train, group_id=group_train)
```

Impact: Learns WavLM scores matter more for short/EXT audio, NeMo TDT for long/DD audio, edit distance consensus is a strong signal. Improved ensemble CER from ~0.272 → ~0.263.

6. Machine Specs & Time

Training was performed across a mix of a local workstation and rented cloud GPUs:

Resource	Local workstation	Rented (NeMo training)	Rented (WavLM training)
CPU	Intel(R) Xeon(R) Platinum 8336C @ 2.30 GHz	(cloud, comparable Xeon class)	(cloud, comparable Xeon class)
GPU	NVIDIA RTX 4090 (24 GB)	NVIDIA RTX 5090 (32 GB)	NVIDIA RTX PRO 6000 (96 GB)
System RAM	1024 GB	62 GB	120 GB
OS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS

Train duration (per single model):

- NeMo Parakeet-TDT-0.6B: **~1-2 hours** (5 epochs, RTX 4090/5090)
- WavLM-Large: **~15-25 hours** (3-5 epochs, RTX PRO 6000)
- Total wall-clock for the 11-model ensemble: **~100-150 GPU hours**
- CatBoost reranker (5-fold): **~30 min** (CPU)

Inference duration:

- Final 11-model ensemble + CatBoost reranker on the competition runtime (NVIDIA A100 80 GB): **~30-56 minutes** for the full test set.

7. Caveats & Known Issues

1. **Inference time:** 11-model ensemble takes ~56 min on A100, tight for competition time limit. Inference order matters — run TDT models first (slow but don't need CTC logprobs), then CTC models (fast with NeMo fast decode ~580 utt/s).
2. **EXT CV bias:** Early local CV for TalkBank data was overly optimistic. Fixed by ensuring proper `child_id` grouping in `StratifiedGroupKFold`. Corrected CV aligned well with LB (offset ≈ -0.008).
3. **WavLM memory:** WavLM-Large requires significant GPU memory. Used gradient accumulation on Pro 6000.
4. **bfloat16:** Inference uses bfloat16 with careful handling of CTC log-probability underflow.
5. **NeMo version:** Requires `nemo_toolkit>=2.0` for TDT decoding support.

8. Tools for Data Preparation / EDA

- **jiwer:** Official CER/WER computation
- **NeMo toolkit:** Model training, TDT/CTC decoding
- **Transformers (HuggingFace):** WavLM backbone
- **CatBoost / LightGBM:** Tree reranker
- **librosa / soundfile:** Audio loading and processing
- **ICE cream (ic):** Debug logging
- **tqdm:** Progress tracking

9. Performance Evaluation Beyond Competition Metric

Hierarchical breakdowns by multiple dimensions:

Dimension	Slices
Data source	DD (DrivenData) vs EXT (TalkBank)
Age group	3-4 years vs 5+ years
Audio duration	Short (<3s) / Medium (3-10s) / Long (>10s)
Word count	1 word / 2-5 words / 6+ words
Combined	DD/3-4, DD/5+, EXT/3-4, EXT/5+

This breakdown revealed the key insight: WavLM excels on short/EXT, NeMo TDT excels on long/DD → drove the heterogeneous ensemble design.

10. Things Tried That Didn't Make Final

Approach	Result
Whisper-small backbone	Vastly inferior to NeMo (0.56 vs 0.30 CER)
Parakeet-TDT-1.1B	Marginal gain over 0.6B, too slow for ensemble
Focal CTC loss	No clear improvement
Pseudo-label training	Mixed results
N-gram LM shallow fusion	Minimal improvement for IPA sequences
MCER (Minimum CER) loss	Experimented but not in final
LoRA fine-tuning	Full fine-tuning worked better
Logits/Prob-level ensemble	Inferior to N-best + tree reranker
Pure MBR ensemble	Good but tree reranker consistently better
Dual char head (vs BPE)	Inferior to dual BPE head

11. Future Improvements

1. **More WavLM variants** with different augmentations — WavLM showed the most complementary performance
2. **IPA-specific language model** for N-best rescoring
3. **Cross-utterance context** — children's speech has contextual cues from surrounding utterances
4. **Self-training / pseudo-labels** on unlabeled TalkBank audio
5. **Larger pre-trained models** (NeMo Parakeet-1.1B or Canary-1B)

12. Simplifications for Faster Inference

Simplification	Speed Gain	Accuracy Cost
3 models (1 WavLM + 1 TDT + 1 CTC)	~4x faster	~0.01-0.02 CER
Remove tree reranker, use N-best rescore	~10% faster	~0.005 CER

Simplification	Speed Gain	Accuracy Cost
Reduce beam_width 10→5	~30% faster	~0.002 CER
CTC-only decoding (skip TDT)	~20% faster	~0.003 CER

A minimal 3-model pipeline with N-best rescore would achieve ~0.27 CER in ~15 min on A100.

Submission History

Ver	Date	Public	Private	Key Change
1	02-16	0.5468	0.5464	Whisper CTC baseline (d_model=384)
2	02-16	0.5620	0.5620	Whisper CTC, full train
3	02-16	0.5621	0.5626	Whisper CTC, lr=1e-4
4	02-24	0.3009	0.3017	NeMo Parakeet-TDT-0.6B + EXT data
5	02-26	0.2941	0.2936	Head lr(1e-4), corpus_level_loss
6	03-01	0.2884	0.2893	New submit framework, ctc_weight=1.0
7	03-03	0.2886	0.2893	Pretrain word→IPA (no gain)
8	03-06	0.2830	0.2835	First ensemble: 4-model MBR

Ver	Date	Public	Private	Key Change
9	03-08	0.2781	0.2778	6-model N-best rescore
10	03-11	0.2754	0.2761	10-model N-best rescore
11	03-16	0.2716	0.2733	5-model optimized pipeline (1 TDT + 4 CTC)
12	03-18	0.2687	0.2705	First tree reranker: 8 models + CatBoost
13	03-19	0.2605	0.2620	WavLM-Large added: 9 heterogeneous models
14	03-23	0.2582	0.2598	10 models (3 WavLM + 7 NeMo)
15	03-25	0.2551	0.2569	10 models, full train (offline→online)
16	04-06	0.2539	0.2559	Final: 11 models (4 WavLM + 7 NeMo) + tree reranker

Evolution Phases

Phase	Submissions	Core Innovation	Relative Improvement
Whisper Baseline	v1→v3	Whisper CTC small model	Baseline (0.547)

Phase	Submissions	Core Innovation	Relative Improvement
NeMo Backbone	v3→v4	Switch to Parakeet-TDT-0.6B	-46.4% (0.562→0.301)
Single Model Tuning	v4→v7	Head lr, corpus loss, EXT data	-4.1% (0.301→0.289)
Ensemble	v7→v11	MBR → N-best rescore	-6.0% (0.289→0.272)
Tree Reranker	v11→v12	CatBoost LambdaRank	-1.1% (0.272→0.269)
Heterogeneous	v12→v16	Add WavLM-Large models	-5.6% (0.269→0.254)
Overall	v1→v16	Full pipeline	-53.6% (0.547→0.254)

Solution by Chenghuige. Code authored with GPT-4 and Claude assistance.